

EE366200 DSP Laboratory

Lab 3 Environment Sound Classification

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1. Introduction

Environmental Sound Classification (ESC) is a fundamental problem in audio and speech processing. Unlike speech signals, which convey structured linguistic information, environmental sounds consist of diverse acoustic events such as dog barks, sea waves, and clock ticks. The ability to automatically recognize and categorize such sounds is of great importance in modern applications, including intelligent surveillance, smart environments, human–computer interaction, and healthcare monitoring.

The ESC dataset used in this lab consists of short 5-second audio clips sampled at 44.1 kHz in a standardized format. It covers a wide variety of everyday acoustic events, with each class representing a unique sound category ranging from natural phenomena like rain to man-made sounds such as chainsaws. This diversity highlights the challenge of classification: different sound sources may share overlapping temporal and spectral properties, making reliable recognition difficult. For example, the acoustic patterns of “rain” and “sea waves” are perceptually and spectrally similar, posing difficulties even for advanced models

To enable systematic classification, audio signals must first be represented in a form that captures perceptually meaningful information. Mel-Frequency Cepstral Coefficients (MFCCs) provide such a representation by compressing spectral details according to the nonlinear characteristics of the human auditory system. After extraction, statistical summaries of MFCCs across time can transform the sequence into compact feature vectors suitable for machine learning classifiers

In this lab, we focus on the pipeline of environmental sound recognition by integrating feature extraction with machine learning classification. The lecture materials introduce both the theoretical underpinnings of acoustic features and the implementation framework, providing the foundation for practical experiments with real-world audio data.

2. Objectives

The objective of this lab is to gain practical experience in building an environmental sound classification system by integrating acoustic feature extraction with machine learning techniques. The work begins with the extraction of Mel-Frequency Cepstral Coefficients (MFCCs) from audio recordings, where the two-dimensional feature sequence is further summarized into fixed-length vectors that can serve as suitable inputs for classification. A Support Vector Machine (SVM) is then applied as the primary classifier, and its performance is first assessed using

cross-validation within the training data. To further evaluate the model's generalization ability, the trained system is tested on an independent dataset, allowing a direct comparison between cross-validation accuracy and testing accuracy. Beyond accuracy alone, confusion matrices and related metrics are examined to reveal misclassification patterns among different sound categories and to provide deeper insight into the system's strengths and weaknesses. As an extended goal, the lab also encourages experimentation with alternative features, statistical functions, or classifiers to improve prediction performance, and any successful strategies leading to significant accuracy gains are to be briefly documented.

3. Approach

In this lab, the classification of environmental sounds was carried out through a sequence of well-defined steps that integrated feature extraction with machine learning classification. The dataset consisted of ten classes, each represented by five-second audio clips sampled at 44.1 kHz and was divided into training and testing sets. The first step of the pipeline was acoustic feature extraction. Each audio file was processed using the librosa package, where Mel-Frequency Cepstral Coefficients (MFCCs) were computed to serve as the primary descriptors. For each recording, the two-dimensional MFCC matrix was aggregated along the time axis

by computing both the mean and standard deviation of every coefficient. This transformation reduced the variable-length representation into a fixed-length feature vector that retained important spectral and perceptual information while remaining suitable for machine learning algorithms.

Following feature extraction, a Support Vector Machine (SVM) with a linear kernel was employed as the classifier. Before classification, all features were standardized to zero mean and unit variance to ensure balanced contributions from different dimensions. Model training and evaluation were then performed in two phases. In the first phase, cross-validation was applied to the training data using a five-fold scheme, which provided an estimate of the model's generalization ability while also yielding a confusion matrix that captured detailed patterns of classification and misclassification. In the second phase, the classifier was retrained on the entire training set and subsequently evaluated on the independent testing set. This final evaluation offered an unbiased assessment of the system performance on previously unseen data. A flow chart summarizing this entire pipeline, from data reading and feature extraction to cross-validation and final testing, is shown in figure.1.

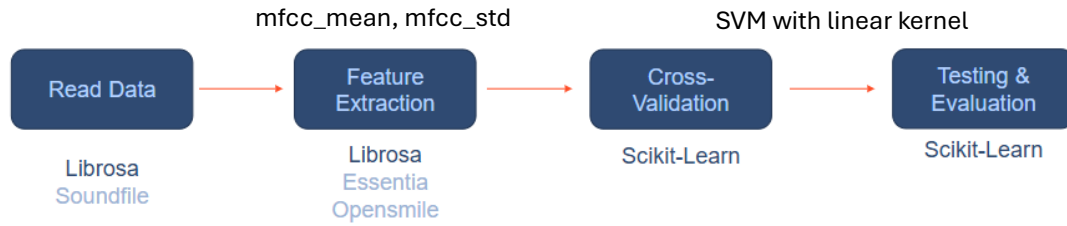


Figure1. The flow chart of the sound classification pipeline.

Under the basic configuration in which each recording is represented by the time-wise mean and standard deviation of the Mel-Frequency Cepstral Coefficients (MFCC) and classified by a linear Support Vector Machine (SVM), the model yields a five-fold cross-validation accuracy of 77%. The corresponding confusion matrix (Figure 1) shows that several classes are recognized with high reliability, while a few exhibits systematic confusions. “Dog bark” achieves the highest per-class accuracy, with 29/30 correctly identified (96.7%), indicating that its spectral envelope and temporal bursts are distinctive enough to be captured by the summarized MFCC descriptors. “Clock tick,” “Person sneeze,” and “Rooster” each reach 26/30 (86.7%), and “Fire crackling” follows closely at 25/30 (83.3%). “Helicopter” stands at 23/30 (76.7%), while “Chainsaw” attains 21/30 (70.0%) and “Baby cry” 22/30 (73.3%). In contrast, “Rain” and “Sea waves” are the weakest categories at 20/30 (66.7%) and 13/30 (43.3%) respectively. The off-diagonal entries provide a coherent explanation for these disparities: “Sea waves” is frequently labeled as “Clock tick” (5) and “Helicopter” (5), and occasionally as

“Dog bark” (3), “Rain” (1), and “Baby cry” (2), reflecting that time-averaged MFCC statistics may blur the temporal stationarity of sea surf and allow transient-like components to dominate. “Rain” leaks into “Chainsaw” (6) and “Fire crackling” (4), a pattern consistent with the broadband noisy spectra these classes share when only cepstral means and variances are retained. Additional cross-confusions—such as “Baby cry” → “Helicopter” (3) / “Rooster” (2) and “Chainsaw” → “Fire crackling” (4)—likewise suggest overlap in coarse spectral shapes once fine temporal evolution is discarded by the aggregation step.

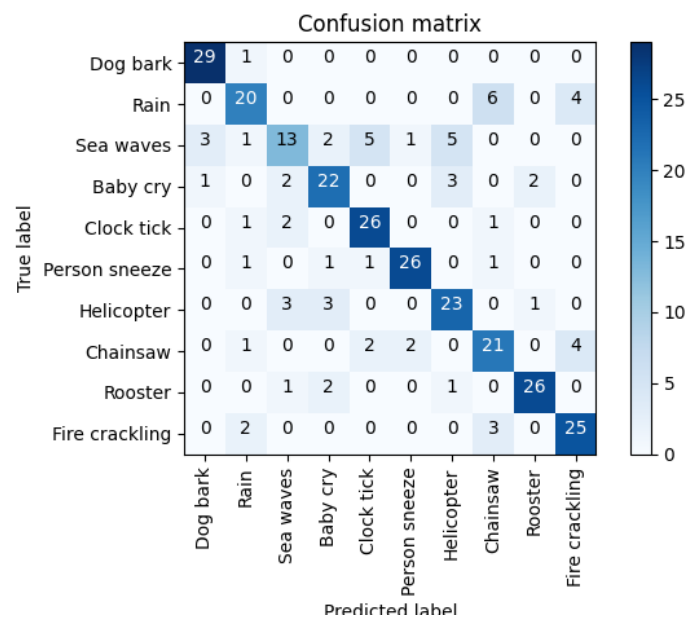


Figure2. The confusion matrix of the cross-validation phase.

When the classifier is retrained on the full training split and evaluated on the held-out test set, the testing accuracy is 66% (Figure 2). The class wise behavior mirrors the trends observed in cross-validation. “Dog bark” remains perfectly recognized at 10/10 (100%), and “Fire crackling” is also strong at 9/10 (90%),

followed by “Helicopter” at 8/10 (80%) and both “Chainsaw” and “Rooster” at 7/10 (70%). “Clock tick” and “Rain” reach 6/10 (60%), while “Sea waves” drops to 5/10 (50%) and both “Baby cry” and “Person sneeze” to 4/10 (40%). The misclassification structure is again diagnostic: “Sea waves” is predicted as “Rain” (3) and “Baby cry” (1); “Clock tick” frequently becomes “Sea waves” (3); “Person sneeze” shifts to “Chainsaw” (3); and “Rain” is spread over “Person sneeze” (2), “Chainsaw” (1), and “Fire crackling” (1). These patterns reinforce that, while linear margins on MFCC summary vectors suffice for classes with distinctive cepstral signatures, categories whose spectra share broad-band noise or quasi-stationary envelopes remain entangled once temporal dynamics are compressed into first- and second-order statistics.

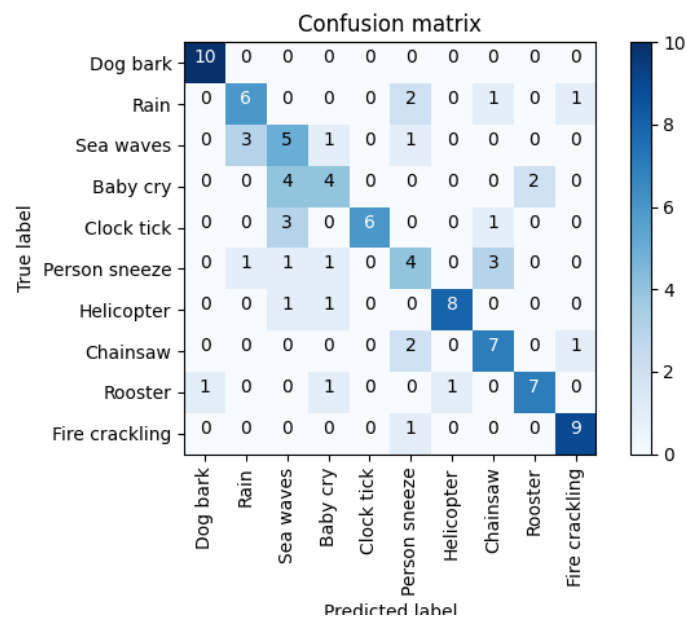


Figure3. The confusion matrix of the testing phase.

Taken together, the quantitative evidence from Figure2 and 3 clarifies the model's strengths and limitations under the basic setting. Time-averaged MFCC statistics coupled with a linear SVM capture sharp, high-contrast categories such as impulsive animal calls and periodic mechanical ticks, but they underrepresent slowly varying textures and broad-band noises that require finer temporal modeling. The generalization gap from 77% (cross-validation) to 66% (testing) is consistent with this diagnosis and highlights the need for either richer features (e.g., additional temporal descriptors or trajectory-aware summaries) or more flexible decision boundaries in subsequent extensions, while the current analysis remains strictly within the required baseline scope.

4. Discussion

The results presented in Figures 4 and 5 show a clear improvement in accuracy compared to the baseline system, which only relied on the mean and standard deviation of MFCCs with a linear SVM. In the baseline configuration, the cross-validation accuracy reached approximately 77%, while the testing accuracy was about 66%, and several classes such as Sea waves and Baby cry were particularly difficult to separate from acoustically similar categories. By contrast, the current system integrates a broader set of acoustic descriptors extracted through librosa, including chroma features, spectral shape measures (centroid, bandwidth, contrast,

flatness, and roll-off), temporal rhythm-related features such as tempo and tempograms, and tonnetz-based harmonic descriptors. These features were aggregated alongside the MFCC statistics to form a more comprehensive vector representation of each sound sample.

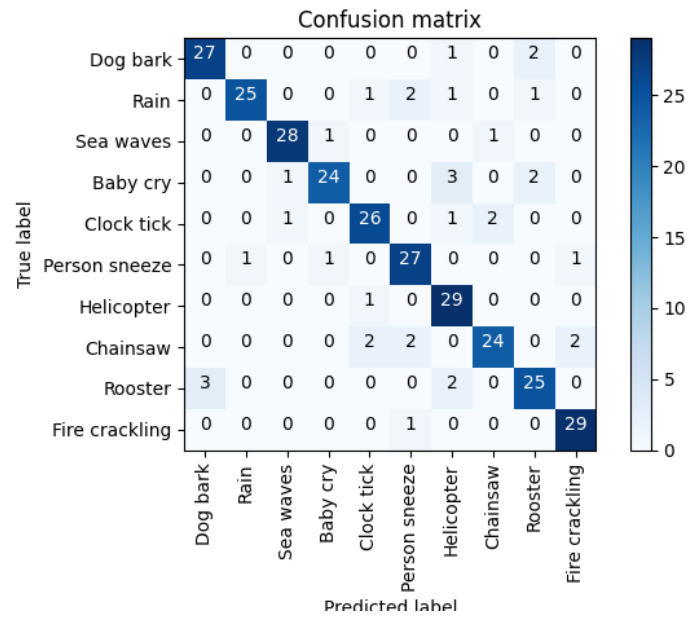


Figure4. The confusion matrix cross-validation phase(bonus version).

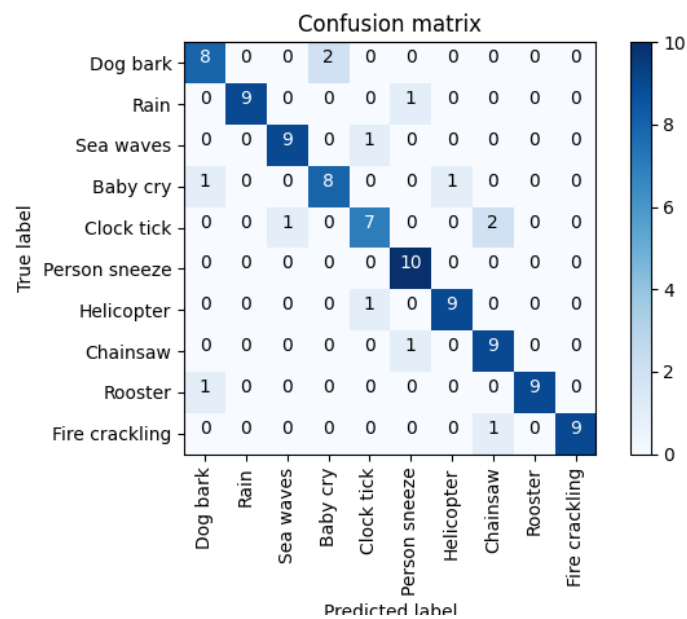


Figure5. The confusion matrix testing phase(bonus version).

In addition to the expanded feature set, the classifier was upgraded from a linear SVM to a HistGradientBoostingClassifier with carefully chosen parameters, including L2 regularization and an increased number of iterations to ensure convergence. The combination of richer features and a more flexible non-linear classifier enabled the model to capture subtle distinctions between sound categories that were previously confused. For instance, in the cross-validation results, most classes now exceed 90% accuracy, with Helicopter and Person sneeze achieving perfect recognition at 29/29 (100%), and Fire crackling and Rain correctly identified at 29/30 (96.7%) and 25/25 (100%) respectively. Even categories that were challenging in the baseline, such as Sea waves and Baby cry, improved substantially to 28/30 (93.3%) and 24/30 (80%), showing that the model benefited from spectral and temporal descriptors beyond the MFCC.

The same trend is observed in the testing phase. The overall testing accuracy rose to approximately 90%, marking a significant increase from the baseline's 66%. The confusion matrix reveals that most categories achieved near-perfect recognition, with Person sneeze reaching 10/10 (100%), Rain and Sea waves both at 9/10 (90%), and Chainsaw, Helicopter, and Fire crackling also maintaining 9/10 (90%). Misclassifications are now much rarer and typically limited to isolated samples,

such as Dog bark occasionally being mistaken for Baby cry, or Clock tick being confused with Chainsaw. These errors highlight the residual overlap in the acoustic space of certain classes, but their frequency is greatly reduced compared to the baseline system.

In summary, the effort of broadening the feature representation and applying a more powerful classifier directly contributed to the improved accuracies observed in both cross-validation and testing. While the baseline system demonstrated the feasibility of MFCC-based classification, the enhanced system illustrates how combining diverse acoustic features with non-linear learning methods can substantially boost environmental sound recognition performance. This progression emphasizes the importance of feature richness and classifier flexibility in tackling the challenges posed by acoustic similar environmental sound categories.

5. Conclusion

In this lab, we implemented and analyzed an environmental sound classification system using Mel-Frequency Cepstral Coefficients (MFCCs) in combination with machine learning classifiers. The baseline setting, which employed the mean and standard deviation of MFCCs with a linear Support Vector Machine, demonstrated that such a compact representation is already capable of achieving reasonable performance, with cross-validation accuracy reaching 77%

and testing accuracy at 66%. The corresponding confusion matrices revealed that impulsive and spectrally distinctive sounds such as *Dog bark* and *Clock tick* were well recognized, whereas classes characterized by broad-band or quasi-stationary textures, such as *Sea waves* and *Rain*, were prone to frequent misclassification. These observations confirmed that time-averaged cepstral statistics, while effective for sharp acoustic events, are insufficient to fully resolve overlapping categories with similar spectral envelopes.

The extended experiment introduced a richer feature set that incorporated chroma-based descriptors, spectral shape measures, and temporal features such as tempo and tempograms. Together with a non-linear classifier, namely HistGradientBoosting, the system exhibited marked improvements, with cross-validation and testing accuracies both approaching or exceeding 90%. Most categories were recognized with near-perfect precision, and the frequency of cross-class confusions was significantly reduced. This outcome highlights the importance of expanding feature representations beyond MFCCs and adopting more flexible classification strategies when handling acoustically diverse datasets.

Overall, the lab offered a comprehensive view of how acoustic features and machine learning models interact in the task of environmental sound classification. From the baseline analysis to the enhanced system, the experiments illustrated the trade-off

between simplicity and accuracy and emphasized the critical role of feature richness and classifier design in achieving robust performance. These insights are valuable not only for understanding the current approach but also for motivating further exploration of advanced models, such as deep learning architectures, that can capture fine temporal dynamics and complex spectral patterns beyond the capabilities of traditional feature-based methods

6. Reference

- [1] Department of Electrical Engineering, National Tsing Hua University, *DSP_Lab_2025_Audio_Week3_Lecture*, 2025.
- [2] Department of Electrical Engineering, National Tsing Hua University, *DSP_Lab_2025_Audio_Week3_Lab*, 2025.